



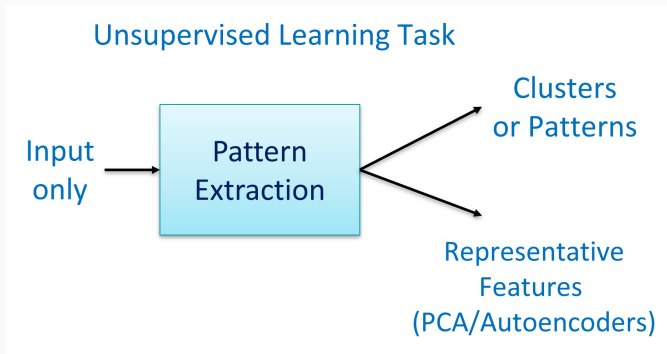
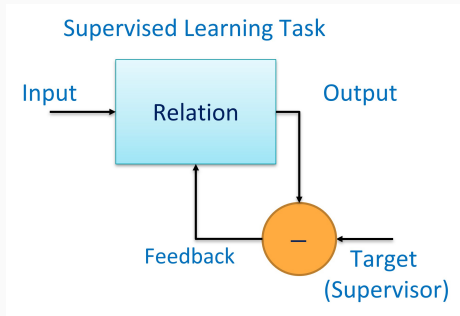
Intro to Reinforcement Learning

DA5400W: FML

P Satya Jayadev

Web Enabled M.Tech,
Indian Institute of Technology Madras

Recall: Supervised and Unsupervised Learning



Motivation: Use Cases where SL and UL fail or are inadequate

- ▶ **Game Playing:** Suppose you want a computer to board games (chess, go, etc.), video games, card games, etc.



Motivation: Use Cases where SL and UL fail or are inadequate

- ▶ **Game Playing:** Suppose you want a computer to board games (chess, go, etc.), video games, card games, etc.
 - SL would require optimal moves labeled for every board or screen configuration
- ▶ **Robotics:** Suppose you want a computer to control a robot to perform various tasks such as walking, picking objects, etc.



Motivation: Use Cases where SL and UL fail or are inadequate

- ▶ **Game Playing:** Suppose you want a computer to board games (chess, go, etc.), video games, card games, etc.
 - SL would require optimal moves labeled for every board or screen configuration
- ▶ **Robotics:** Suppose you want a computer to control a robot to perform various tasks such as walking, picking objects, etc.
 - SL needs ground truth for correct motion or action in every configuration
- ▶ **Autonomous Driving:** Suppose you want build a car that autonomously responds to all driving scenarios



Motivation: Use Cases where SL and UL fail or are inadequate

- ▶ **Game Playing:** Suppose you want a computer to board games (chess, go, etc.), video games, card games, etc.
 - SL would require optimal moves labeled for every board or screen configuration
- ▶ **Robotics:** Suppose you want a computer to control a robot to perform various tasks such as walking, picking objects, etc.
 - SL needs ground truth for correct motion or action in every configuration
- ▶ **Autonomous Driving:** Suppose you want build a car that autonomously responds to all driving scenarios
 - SL needs labeled data for each driving scenario
- ▶ **Optimal Control:** Suppose you want a control system that continuously optimizes control decisions based on changing system dynamics and goals



Motivation: Use Cases where SL and UL fail or are inadequate

- ▶ **Game Playing:** Suppose you want a computer to board games (chess, go, etc.), video games, card games, etc.
 - SL would require optimal moves labeled for every board or screen configuration
- ▶ **Robotics:** Suppose you want a computer to control a robot to perform various tasks such as walking, picking objects, etc.
 - SL needs ground truth for correct motion or action in every configuration
- ▶ **Autonomous Driving:** Suppose you want build a car that autonomously responds to all driving scenarios
 - SL needs labeled data for each driving scenario
- ▶ **Optimal Control:** Suppose you want a control system that continuously optimizes control decisions based on changing system dynamics and goals
 - SL would require labeled optimal control actions under all conditions



Motivation: General Scenarios where SL and UL fail or are inadequate

Scenario	Why SL/UL Fail
Sequential Decision Making: In many real-world problems, decisions are interdependent	SL/UL assume independent examples and cannot model sequential dependencies
Delayed or Sparse Feedback: Sometimes the response to an input does not come immediately	SL requires immediate labels; can't handle delayed response or feedback
Interactive Environments: Environment responds dynamically to the learner's actions	SL or UL are passive; they cannot adapt to evolving environments
No Ground Truth Labels: Desired output or optimal action is unknown or hard to specify	SL can't train without labels; UL can only group or reduce data.
Need for Exploration: Learner must try unknown actions to discover better strategies	SL/UL only work on seen data; they cannot explore



Reinforcement Learning: Learning by Interaction

- ▶ Similar to how humans learn to ride a bicycle or ride a car
- ▶ A learning **agent** interacts with an **environment** through a sequence of decisions or actions
- ▶ At each step:
 - Agent observes the environment's **state**
 - Takes an **action**
 - Receives feedback called **reward**
 - Environment changes state
- ▶ **Learning Goal:** To learn a strategy called **policy** that maximizes cumulative reward called **return**



Supervised Learning vs Reinforcement Learning

Aspect	Supervised Learning	Reinforcement Learning
Training Data	Input-Output Pairs	State Observations and rewards
Learning Signal	Target output	Reward
Post Training Objective	To make best predictions	To take optimal actions
Math Objective	Minimize Error in prediction	Maximize Long-term Reward gained

